MOHAMMAD ABU-SHELBAIA

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SUMMARY

Graduated with excellence from Birzeit University with a degree in Computer Engineering. Currently working as a Teaching and Research Assistant at the same institution. Experienced in full-stack web development. Proficient in Python, with a strong focus on automation, web scraping, and backend services.

SKILLS

Languages: Python, LaTeX, Markdown, Bash, Assembly, Java, C, SQL, HTML, CSS

Technologies: Git, FastAPI, Flask, Asyncio, Django, Tailwind CSS, Docker, PostgreSQL, Arduino, Nginx, Gunicorn

EXPERIENCE

9/2024 – Present
Teaching And Research Assistant — Birzeit University
Assisted in delivering laboratory sessions on microcontroller programming and embedded systems. Provided mentorship and support to students, evaluated assignments, and contributed to the preparation of lab materials.

7/2024 – 12/2024
Backend Developer — Silal
Contributed to the development and maintenance of a data processing pipeline for managing and analyzing product data. Wrote unit tests to ensure the reliability of Flask API endpoints.

3/2024 – 7/2024
Backend Developer Intern — Silal
Built a full-stack bookstore application using Django and Tailwind CSS, implementing core functionality and an intuitive user interface. Deployed the application with Docker Compose, using Nginx, Gunicorn, and PostgreSQL for a robust and production-ready setup.

PROJECTS

Education, VitePress
Embedded, Android
ECE Lab Manuals — birzeit.me
Built and maintain open, step-by-step lab manuals for three Birzeit University ECE labs: ARM Cortex-M4, STM32, and Android with Spring Boot.

Embedded Systems
PIC16f877A-Assembly
PIC16f877A Calculator — github.com
Developed a functional calculator on a PIC16f877A microcontroller, showcasing skills in embedded systems design and assembly language programming.

Security, Cryptography
CBC, Docker
Padding Oracle Attack — github.com
Implemented a practical padding oracle attack against CBC mode with PKCS#5 padding, recovering the plaintext byte by byte, and packaged it to run in Docker.

Web Scrapping
Python, iCal
Course Schedule — github.com
Developed a Python script to automate the extraction of weekly schedules from my university website and convert them into recurring iCal events. This project aimed to improve personal organization and time management by eliminating manual schedule data entry.

EDUCATION

9/2020 - 7/2024
Birzeit University — Bachelor
Computer Enginering
`Cumulative Average: 87%`

28th July 2024
Udacity — Nanodegree Certificate
Data Analyst
`Certificate Link`

LANGUAGES

English - C2 (CEFR), **Arabic** - Native